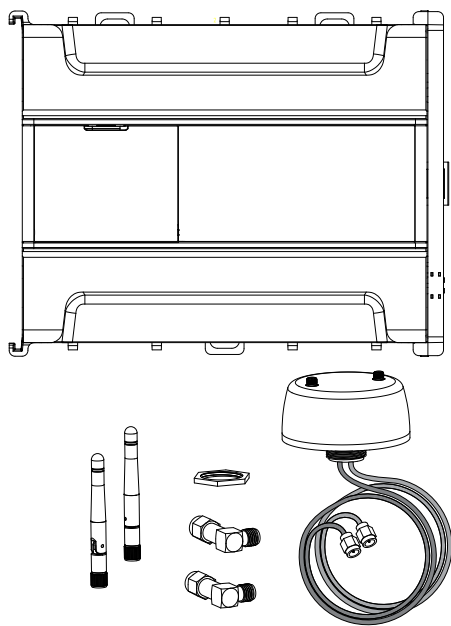


Advanced Controller

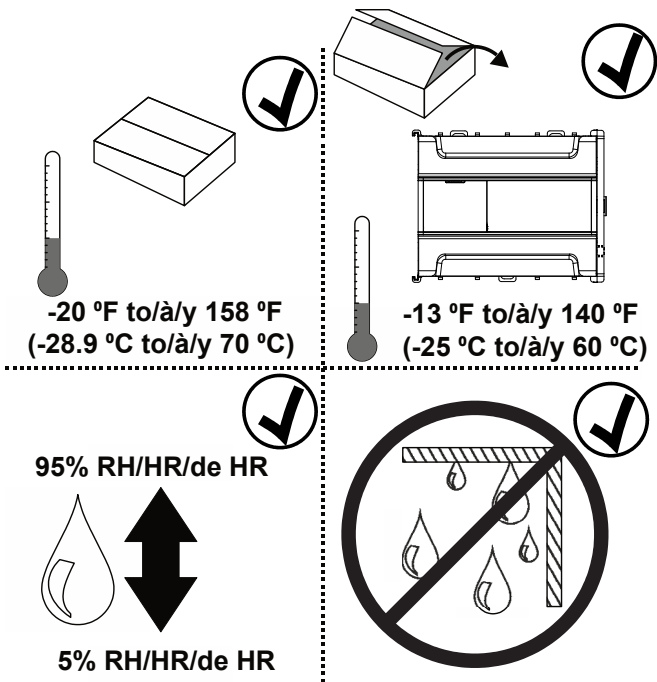
MOUNTING INSTRUCTIONS

1. CONTENTS OF SHIPMENT

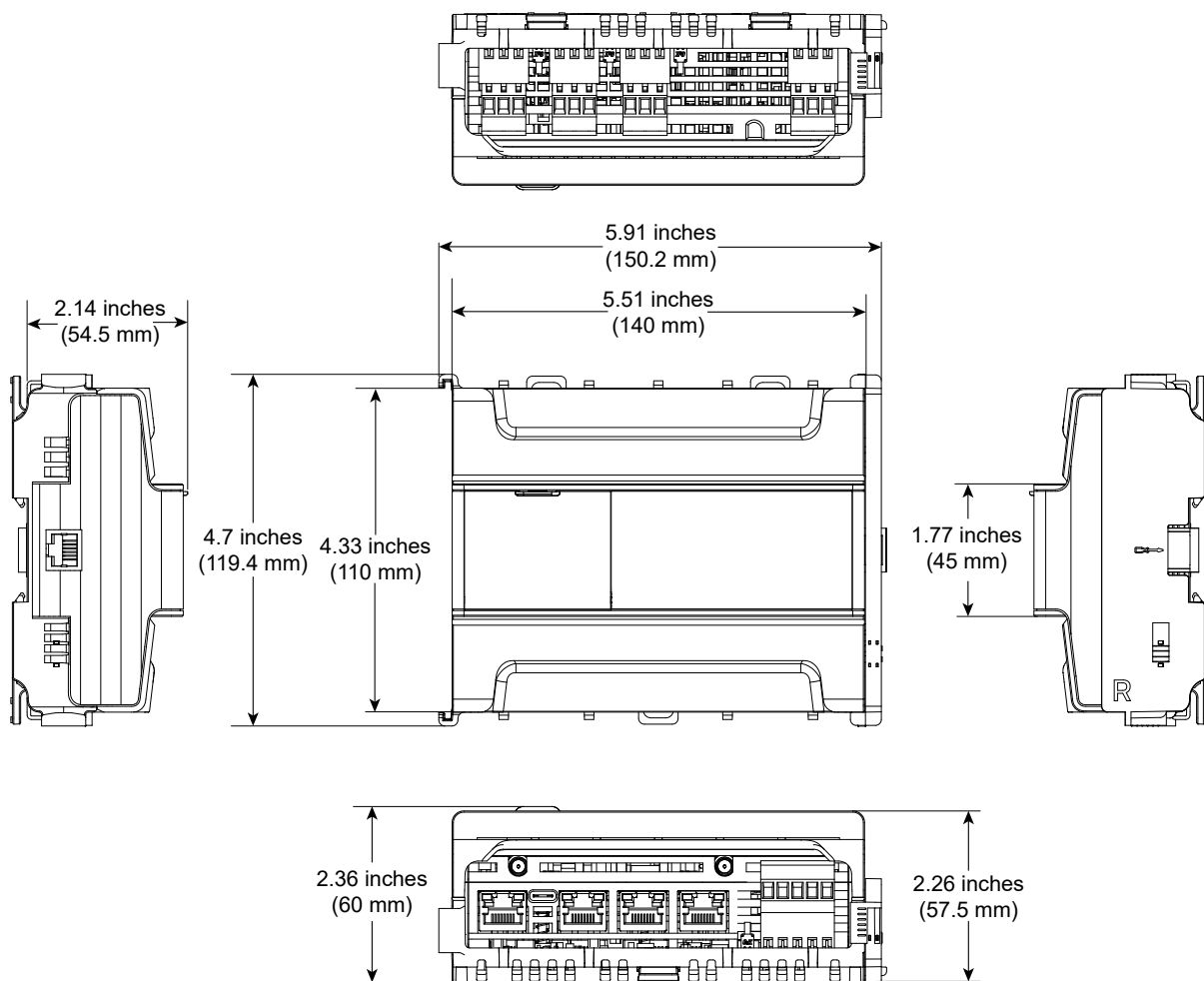
- Controller with Protective Cover: 1 Qty
- Safety Sheet: 1 Qty
- Antenna Kit: 1Qty (Only supplied on Wi-Fi models)



2. STORAGE AND OPERATION



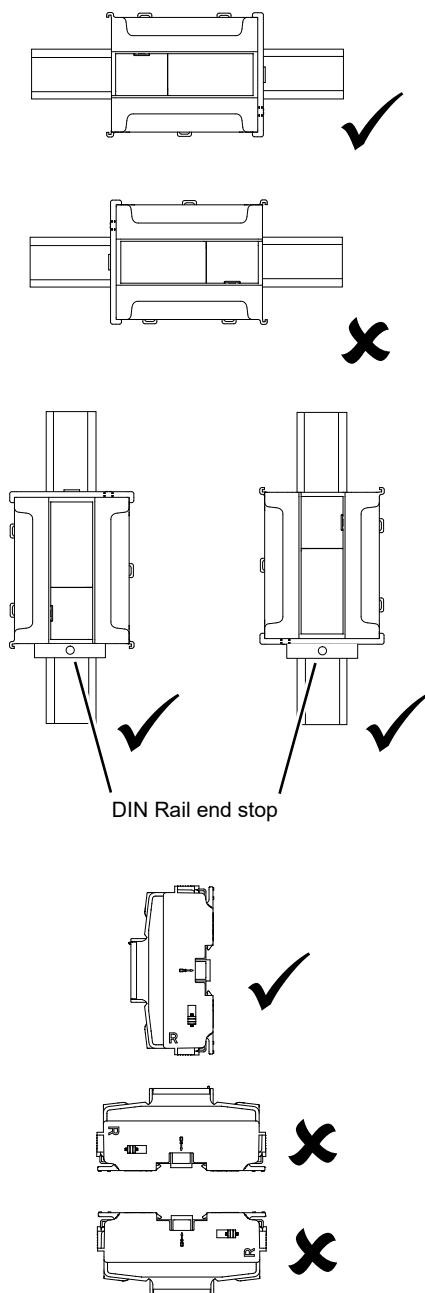
3. DIMENSIONS



NOTE: All the dimensions are in inches (mm). For wiring adapter dimension drawing please refer the Comfortpoint Niagara Advanced Plant Controller Product Datasheet -31-00583 and Optimizer Advanced Controller Product Datasheet -31-00631.

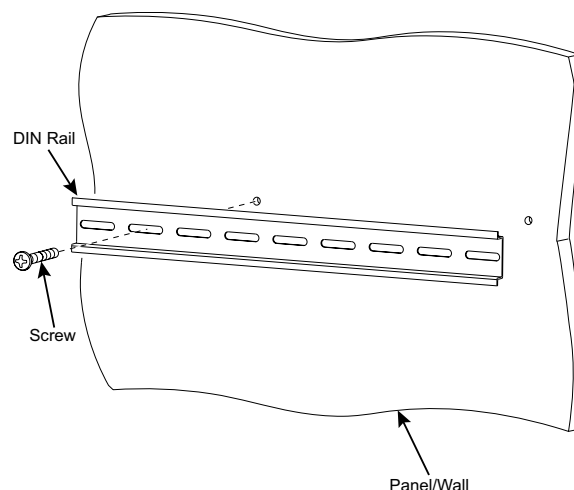
4. MOUNTING PRECAUTIONS

The controllers and wiring adapter support both horizontal and vertical DIN rail mounting on a vertical surface. Do not mount on a horizontal surface.

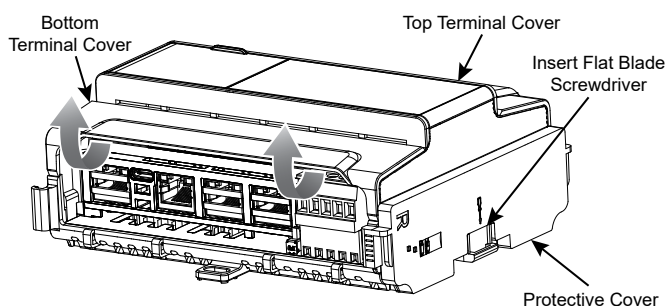


5. MOUNTING THE CONTROLLER ON A WALL OR PANEL DIN RAIL

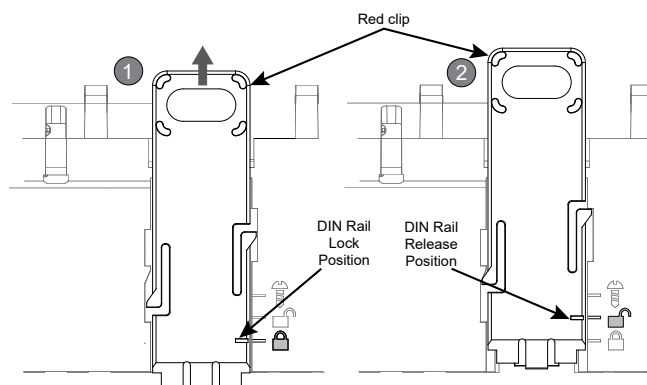
1. Mount the DIN rail on the wall or panel by using screws.



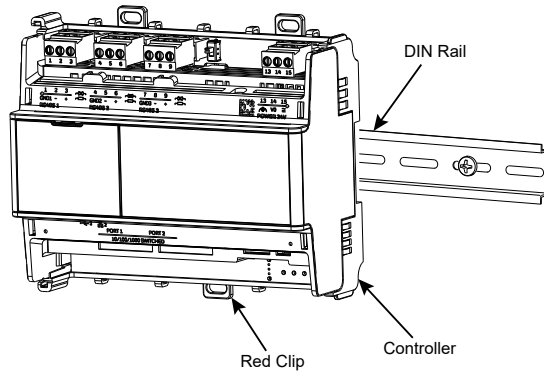
2. Remove the bottom Terminal cover from the controller.



3. Remove the top Terminal Cover from the controller.
4. If IO modules need to be mounted and connected to the Advanced controller using the touch flake connections remove the factory-installed Protective Cover by inserting a flat blade screwdriver into the slot as shown in the above image and push down. If not, keep the Protective Cover attached to the Controller to safeguard the touch flakes.
5. Extend all red clips to the unlock position as shown in the below figure.



6. Hold the controller in an orientation such that the red clip is facing downwards and towards the DIN rail.

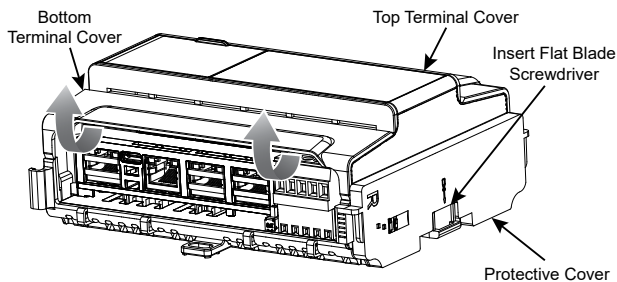


7. Mount the controller onto the DIN rail and push all red clips in to secure it in place.
8. Connect the terminal wires, and Ethernet cables.
9. Replace the terminal covers.
10. If IO modules will not be connected to the Advanced controller using the touch flake connections, please mount and use the Protective Cover.

NOTE: The Protective Cover must be fitted to protect the touch flakes on the side of the controller or IO module. The Advanced controller includes the terminating resistor for the end of the IO BUS, ensure the Protective Cover is always installed at the end of the IO BUS.

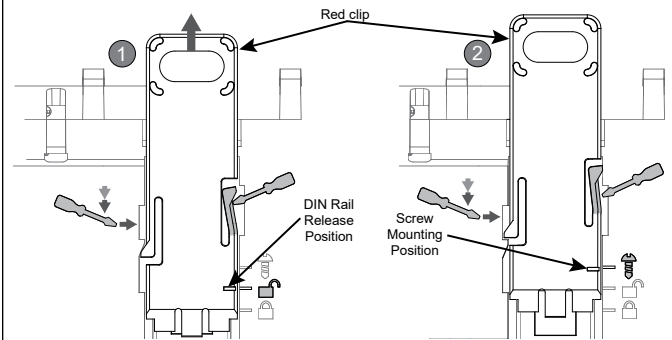
6. MOUNTING THE CONTROLLER ON A WALL/PANEL USING SCREWS

1. Remove the bottom Terminal cover from the controller.



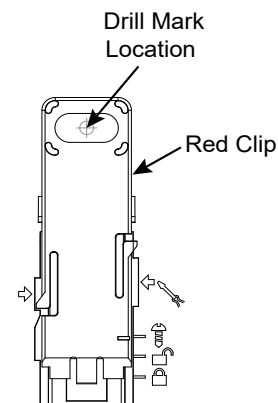
2. Remove the top Terminal cover as described in the previous step.
3. If IO modules need to be mounted and connected to the Advanced controller using the touch flake connections, remove the factory-installed Protective Cover by inserting a flat blade screwdriver into the slot as shown in the above image and push down. If not, keep the Protective Cover attached to the Controller to safeguard the touch flakes.

4. Extend all red clips to the screw mounting position by inserting a flat blade screwdriver at the marked location and moving the notch from the lower to the upper slot, as shown in the figure below.



NOTE: Repeat step 5 of Mounting the controller on a wall or panel using Din rail in case the red clip is in the lock position.

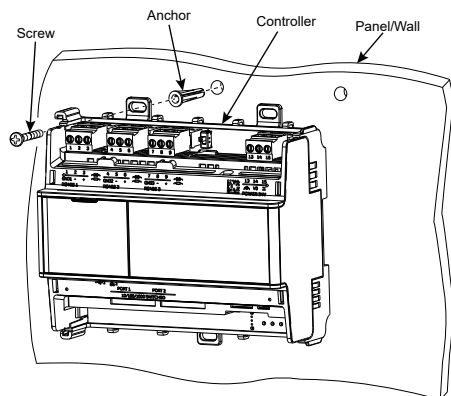
5. Hold the controller along the wall and mark three drilling locations through the red clip slots, as shown in the below figure.



6. Remove the controller from the wall and drill three holes at the marked locations.
7. Insert anchors into the three mounting screw holes.
8. Place the controller on the wall or panel so that the holes are aligned.
9. Insert the screws into the topside holes first and fasten them with a screwdriver.

NOTE: It is recommended to use the 6-18 1" pan head Phillips tapping screws.

10. Insert the screws into the bottom hole and fasten them with a screwdriver.



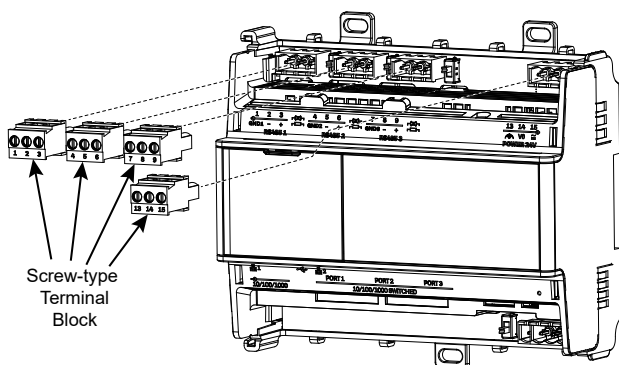
11. Connect the terminal wires, and Ethernet cables.
12. Replace the terminal covers.
13. If IO modules will not be connected to the Advanced controller using the touch flake connections, please mount and use the Protective Cover.

7. TERMINAL BLOCKS

The controller has the screw type terminal blocks by default, factory-installed, and these are removable. Additionally, the controller supports push-in type terminal blocks that are available for purchasing.

Installing the Push-In Terminal Blocks

1. Remove the screw-type terminal blocks from the controller by gently pulling them out manually.



2. Install the Push-in terminal blocks onto the controller.
3. Remove the insulation 0.19 inches (5 mm) from the wires' end by using a stripping tool.
4. Refer to Optimizer Advanced Installation Instruction and Commissioning Guide - 31-00584 for wiring connections and insert the wires into the terminal blocks.

8. ANTENNA

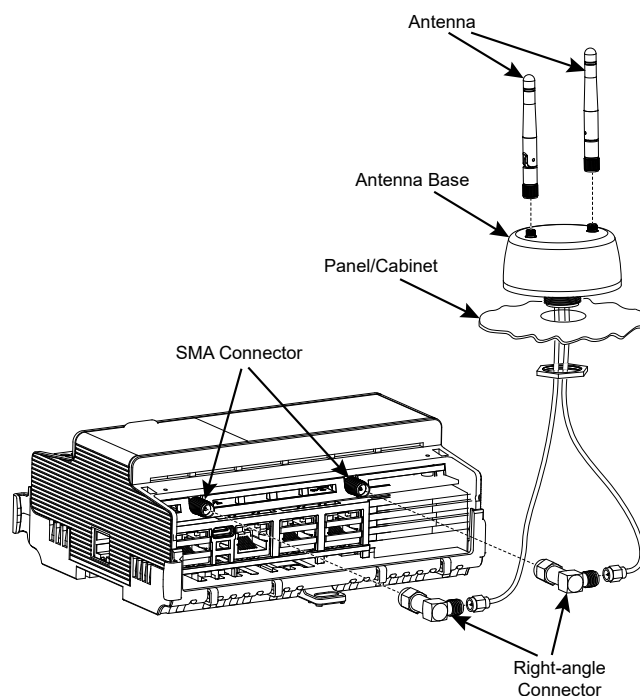
An antenna kit is supplied with the Advanced Wi-Fi controllers only.

The antenna is for wireless communication with other devices via Wi-Fi and Bluetooth®.

The antenna kit includes an antenna base with a 4.92 ft (1.5 m) cable that allows both antennas to be installed remotely if extended range is required or if the controller is mounted in a metal cabinet/enclosure.

Installing the Antennas Remotely (Metal Enclosures)

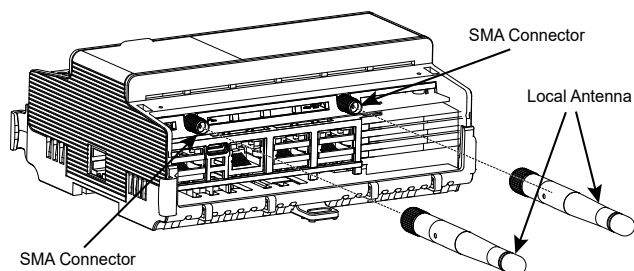
1. When the connection of the antenna wires exceeds the maximum cable bend radius (causes signal loss), connect the 90 degree connectors to the SMA connectors on the controller. Tighten the 90 degree connectors using a torque wrench to the range from 0.57 to 0.9 Nm.
2. Find a location (within 4.92 ft (1.5 m) from the controller) where the signal reception is good and mark the place. Refer to the "9. Antenna placement Remote location" on page 6.
3. Drill a 1 in (25 mm) hole to install the antenna base in the marked location.
4. Safely route the antenna base wire through the drilled hole and to the controller.
5. Insert the Antenna Base into the drilled hole and insert the nut on the other side of the panel/cabinet. Tighten the nut.



6. Connect the ends of the antenna base wires to the 90 degree connectors.
7. Install the antennas on the antenna base.

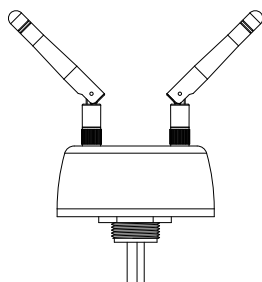
Installing the Local Antennas Directly to the Controller (Non-Metal Enclosure)

1. Insert the local antennas into the SMA connectors.
2. Tighten by turning the base of the antenna clockwise.



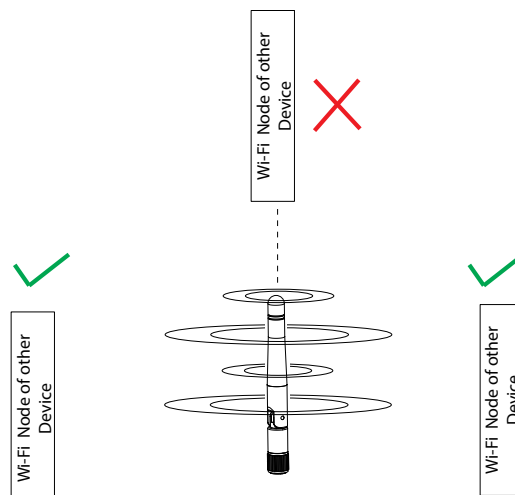
9. ANTENNA PLACEMENT REMOTE LOCATION

1. Place and align the antenna so that the Wi-Fi signal does not interfere with many obstacles.
2. It is recommended to align the antenna angle at 45° to each other for a better signal.

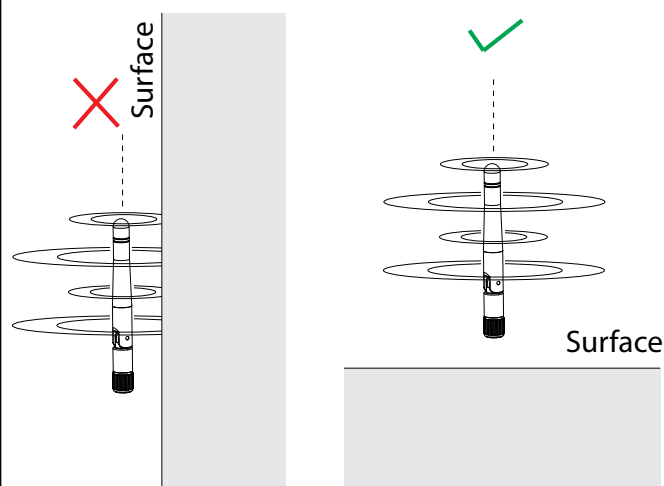


3. Obstacles such as metal and reinforced concrete with steel filter ~10-90% of the Wi-Fi signal. Other obstacles include drywall, wood, and glass filter ~0-10% of the Wi-Fi signal.

4. The antenna has a strong signal transmission/reception from the side. See the figure below. Position the sides of the antenna towards the device it is communicating with other device Wi-Fi Node.



5. Keep the antenna away from devices that cause wireless interference, such as other Wi-Fi networks and telecommunications equipment.
6. Ensure that the antenna is installed perpendicular to the surface for a good Wi-Fi signal.

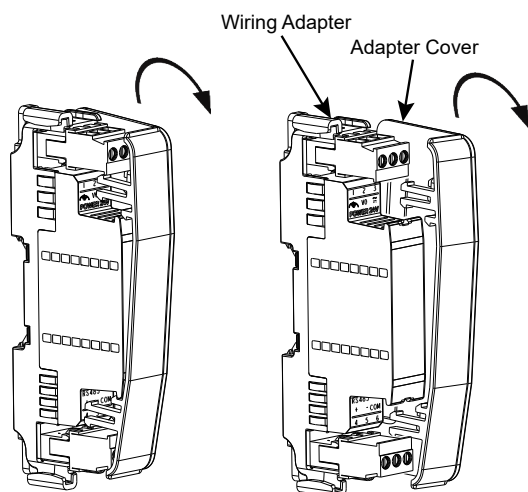


7. Do not install the antenna close to the floor.
8. The Local antenna can be used when the controller is installed in a plastic cabinet/enclosure.

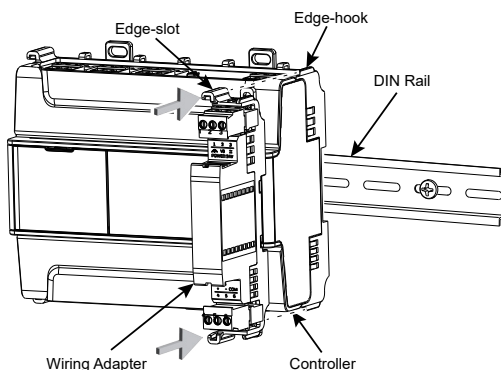
10. MOUNTING THE WIRING ADAPTER ON THE DIN RAIL

NOTE: Use the Wiring Adapter to extend 24 VAC/VDC power and comms to the next DIN rail or to 24 VAC/VDC power and comms to a remote IO module.

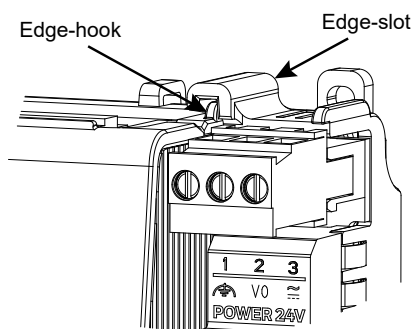
1. Remove the adapter cover by pulling out the cover along the radius arrow as shown in the below image.



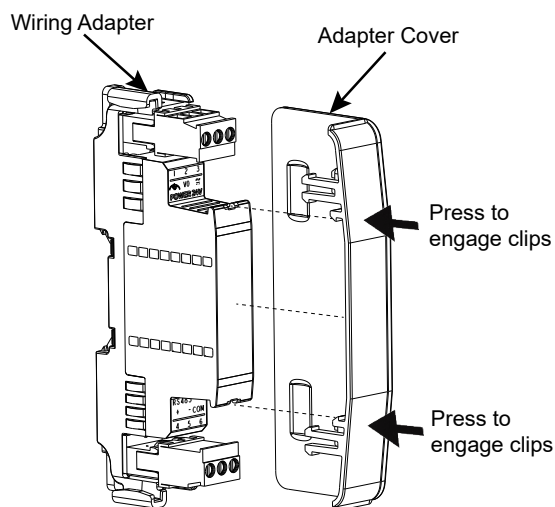
2. Extend the red clips on the wiring adapter to the last position.
3. Hold the wiring adapter in the position as shown below.



4. Mount the adapter in a way that the edge-slot of the adapter aligns with the axis of the edge-hook of the controller to attach and secure them together.

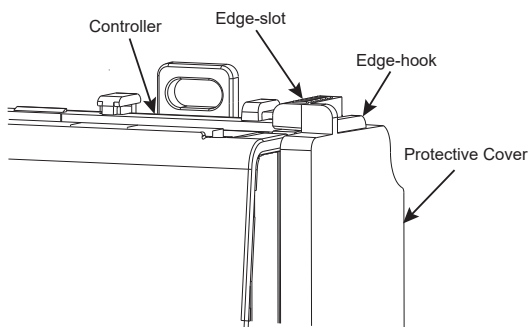
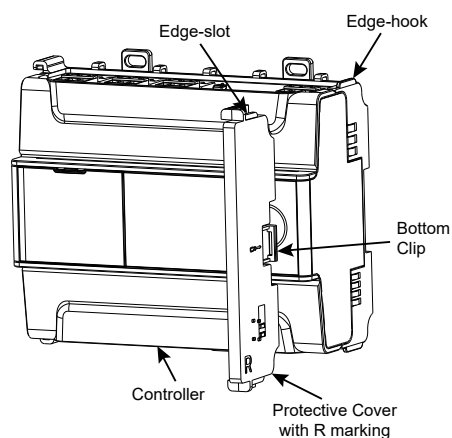


5. Mount the adapter cover on the wiring adapter by firmly pushing it against the axis as shown in the illustration below. There should be no space between the cover and the adapter. In addition to providing end-of-line termination for the RS485 bus, the adapter cover protects the touch flakes.



11 MOUNTING THE PROTECTIVE END COVER ON THE CONTROLLER

1. Hold the Protective end Cover (marked with R), which has the terminating resistor in a way that the edge-slot of the cover aligns with the axis of the edge-hook of the controller, and slide along till the bottom clip is locked with the IO module.



12. REMOVING THE CONTROLLER FROM THE DIN RAIL

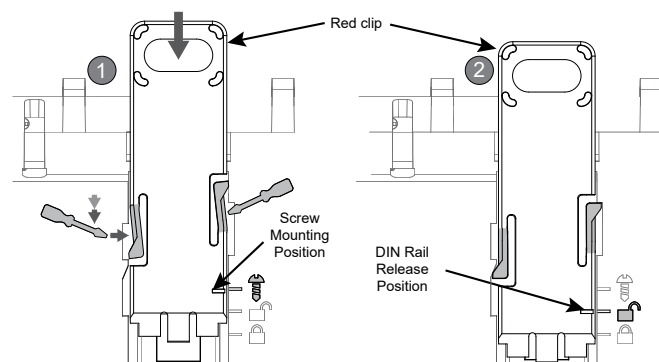
1. Switch off the power supply.
2. Remove the Terminal Covers and Protective Cover if applicable.
3. Disconnect all the controller's communication and power supply terminals.
4. Hold the controller with one hand and insert a flat blade screwdriver into the red clip slot.
5. Pull the red clip downward by using the flat blade screwdriver.
6. After the controller is released from the DIN rail, dis-engage the top two controller clip, slightly tilt the controller on the horizontal axis, lift up and detach the controller from the DIN rail.

13. REMOVING THE CONTROLLER FROM THE WALL OR PANEL

1. Switch off the power supply. Remove the Terminal Covers and Protective Cover if applicable.
2. Remove the Terminal Covers and Protective Cover if applicable.
3. Disconnect all the controller's communication, and power supply terminals.
4. Unscrew the bottom side screws first by using a screwdriver.
5. Hold the controller with one hand, unscrew the top side screws using a screwdriver.

14. SCENARIO FOR RED CLIP

1. If the user wants to attach the controller to a wall or panel using screws, push all red clips down to the unlock mounting position by inserting the flat blade screwdriver at a defined point, as shown in the figure below.



SPECIFICATIONS OF THE CONTROLLER

POWER CONSUMPTION				
CONTROLLER	POWER CONSUMPTION		CURRENT CONSUMPTION	
	24 VAC	24 VDC	24 VAC	24 VDC
N-ADV-134-H	Max. 36 VA	Max. 13 W	1500 mA	540 mA
N-ADV-133-H-BWA N-ADV-133-H-BWE N-ADV-133-H-BWW	Max. 38 VA	Max. 14 W	1600 mA	580 mA
N-ADV-133-H	Max. 34 VA	Max. 12.5 W	1420 mA	520 mA
N-ADV-133-H-B	Max. 34 VA	Max. 12.5 W	1420 mA	520 mA
N-ADV-112-H	Max. 34 VA	Max. 12.5 W	1420 mA	520 mA

ELECTRICAL	
PARAMETER	SPECIFICATION
Operating Voltage (AC)	24 VAC (+/-20%) = 19 to 29 VAC 50/60 Hz
Operating Voltage (DC)	24 VDC (+/-20%) = 20 to 30 VDC
Overvoltage Protection	Protected against overvoltage of max. 29 VAC or 40 VDC. Terminals protected against short-circuiting.

OPERATIONAL ENVIRONMENT	
PARAMETER	SPECIFICATION
Storage Temperature	-20 to 158 °F (-28.9 to +70 °C)
Operating temperature	-13 to 140 °F (-25 to 60 °C)
Humidity	5 to 95 % relative humidity (non-condensing)
Vibration Under Operation	0.024" double amplitude (2 to 30 Hz), 0.6 g (30 to 300 Hz)
Dust, Vibration	According to EN60730-1
Protection	IP20 with optional terminal covers
Altitude	13123 ft (4000m)

STANDARDS AND APPROVALS

Product Standards	UL60730-1, UL60730-2-9, UL916, IEC/EN60730-1, Energy Management Equipment, IEC/EN60730-2-9, CAN/CSA-E60730-1:02, IEC/EN61326-1, and IEC/EN61010.
Certification	UL60730-1, UL916, CE, BTL B-BC, BACnet™ Standard 135 version 1.14, ISO 16484-5, FCC Part15, WEEE, C-tick RCM, Subpart B, CAN ICES-3 (B)/NMB-3(B), RCM, AMEV AS-B, KBOB, EAC, RoHS II, Ethernet Protocol version IEEE802.3, EN-1434-3, EN-13757-3, and ISA/IEC 62443-4-2 SL2 Certified, NBTC, IMDA, KC, SRRC, NTC, IFT/NOM, and NOM.
Shock Protection	SELV
Pollution Class	IP20
Software Class	Class B

IMPORTANT: Keep mains power supply and loads cables separate from signal wiring!

STANDARDS AND APPROVALS



WARNING

Electrical Shock Hazard.

Can cause severe injury, death, or property damage. Disconnect the power supply before beginning installation to prevent electrical shock and equipment damage. More than one power supply may have to be disconnected.

It is prohibited to connect any of the RJ45 sockets of the Advanced controller to a so-called PoE-enabled device ("Power over Ethernet").



CAUTION/MISE EN GARDE/PRECAUCIÓN

To reduce the risk of fire or electric shock, do not interconnect the outputs of different Class 2 circuits.

WEEE



WEEE Directive 2012/19/EC Waste

Electrical and Electronic Equipment Directive

- At the end of the product life, dispose of the packaging and product in an appropriate recycling center.
- Do not dispose of the device with the usual domestic refuse.
- Do not burn the device.

FCC REGULATION

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause interference.
2. This device must accept any interference, including interference that may cause undesired operation of the device.

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

However, there is no guarantee that interference will not occur in a particular installation.

Suppose this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on. In that case, the user is encouraged to try to correct the interference by one or more of the following measures:

1. Reorient or relocate the receiving antenna.
2. Increase the separation between equipment and receiver.
3. Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
4. Consult the dealer or an experienced radio/TV technician for help.

CANADIAN REGULATORY STATEMENT

This device complies with Industry Canada license exempt RSS standard(s). Operation is subject to the following two conditions:

1. This device may not cause interference.
2. This device must accept any interference, including interference that may cause undesired operation of the device.

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes :

1. l'appareil ne doit pas produire de brouillage.
2. l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

CE Statement: The WLAN function for this device is restricted to indoor use only when operating in the 5150 to 5350 MHz frequency range.



BE	BG	CZ	DK	DE	EE	IE	EL	ES	FR	HR	IT	CY
LV	LT	LU	HU	MT	NL	AT	PL	PT	RO	SI	SK	FI
SE	UK (NI)	TR	NO	CH	IS	LI						

EMF Statement: To comply with the RF exposure requirement, a separation distance of 20 cm between the device and the human should be maintained.

Déclaration d'exposition Attention: Cet émetteur doit être installé pour fournir une distance de séparation d'au moins 20 cm de toute personne.

THAILAND REGULATORY STATEMENT

This device is exempted to possess license, user license, or radiocommunication station license as per NBTC notification regarding radiocommunication equipment and radiocommunication station has been exempted for license according to radio communication act B.E.2498.



SINGAPORE REGULATORY STATEMENT

This device complies with Industry Singapore license exempt as per the IMDA (Infocomm Media Development Authority).

Complies with
IMDA Standards
DA107974

STANDARDS AND APPROVALS

RESTRICTIONS IN THE 5 GHZ BAND

Within the 5.15 to 5.25 GHz band, UNII devices will be restricted to indoor operations to reduce any potential for harmful interference to co-channel Mobile Satellite System (MSS) operations.

RESTRICTIONS DANS LA BANDE DE 5 GHZ

Dans la bande de 5,15 à 5,25 GHz, les appareils UNII seront restreints aux opérations intérieures pour réduire toute possibilité d'interférence pouvant nuire aux opérations du Système satellite mobile dans le même canal(MSS).

Professional Installation Warning

- This device must be professionally installed, this should be noted on grantee.
- To maintain compliance, only the antenna types that have been tested shall be used.
- This device requires a significant technology engineering expertise towards understanding of the tools and relevant technology, not readily available to average consumer. Only a person professionally trained in the technology is competent.
- This radio transmitter IC:1609A-ADVBLEWIFI has been approved by Innovation, Science and Economic Development Canada to operate with the antenna types listed below, with the maximum permissible gain indicated. Antenna types not included in this list that have a gain greater than the maximum gain indicated for any type listed are strictly prohibited for use with this device.

CHINA HAZARDOUS SUBSTANCE TABLE

“产品中有害物质的名称及含有的信息表” List of Hazardous Substances in the Product and Relevant Information”						
部件名称 Component Name	有害物质 Hazardous Substances					
	铅 (Pb) Lead (Pb)	汞 (Hg) Mercury (Hg)	镉 (Cd) Cadmium (Cd)	六价铬 (Cr6+) Chromium VI Compounds (Cr6+)	多溴联苯 (PBB) Polybrominated Biphenyls (PBB)	多溴二苯醚 (PBDE) Polybrominated Diphenyl Ethers (PBDE)
PCBA	X	○	○	○	○	○

“产品中有害物质的名称及含有的信息表” List of Hazardous Substances in the Product and Relevant Information”				
部件名称 Component Name	有害物质 Hazardous Substances			
	邻苯二甲酸二(2-乙基)己酯 (DEHP) Bis (2- ethylhexyl phthalate (DEHP)	邻苯二甲酸丁苄酯 (BBP) Benzyl phthalate (BBP)	邻苯二甲酸二正丁酯 (DBP) Dibutylphthalate (DBP)	邻苯二甲酸二异丁酯 (DIBP) Diisobutyl phthalate (DIBP)
PCBA	○	○	○	○

本表格依据 SJ/T 11364 的规定编制。

This table is prepared in accordance with the provisions of SJ/T 11364.

○：表示该有害物质在该部件所有均质材料中的含量均在 GB 26572 规定的限量要求以下。

Indicates that said hazardous substance contained in all of the homogeneous materials for this part is below the limit requirement of GB 26572.

X：表示该有害物质至少在该部件的某一均质材料中的含量超出 GB 26572 规定的限量要求。

Indicates that said hazardous substance contained in all of the homogeneous materials for this part is above the limit requirement of GB 26572.

环保使用年限 EPUP	10					
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未列入表内的其他部件，皆不含任何超出限量要求的限制使用物质

All other components, not listed in the table, do not contain restricted substances above the threshold level

NOTE

This RoHS certification is only applicable for the controller N-ADV-112-H

REFERENCE TECHNICAL LITERATURE

TITLE	LITERATURE NUMBER
COMFORTPOINT_NIAGARA_ADVANCED PLANT CONTROLLER_PRODUCT DATASHEET	31-00583
OPTIMIZER ADVANCED CONTROLLER PRODUCT DATASHEET	31-00631
OPTIMIZER ADVANCED INSTALLATION INSTRUCTION AND COMMISSIONING GUIDE	31-00584

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31-00553-03 | Rev. 04-26

